User Manual

Group 44

This guide explains how to use the command-line Personal Information Manager (PIM) to keep notes, tasks, events and contacts, search them, and save them to a file. PIM never saves by itself: use `save` to keep your records.

1 Start and exit

Install JDK 17 or later. In the PIM folder, build once and start the program:

```
./scripts/build.sh
java -jar build/pim.jar
```

PIM shows a welcome message and the `pim>` prompt (Figure 1). Type `help` for the command list (Figure 2). Type `exit` to leave; if there are unsaved changes, PIM asks you to confirm.

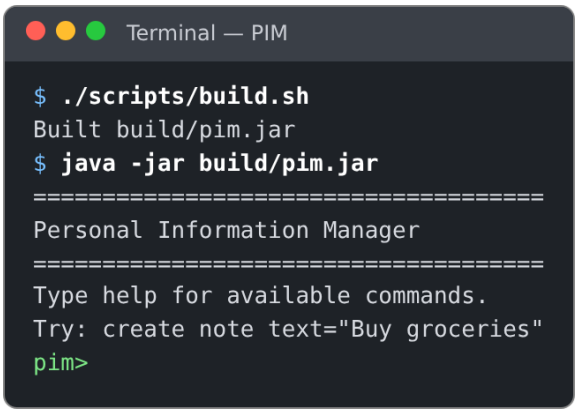


Figure 1: Building PIM and starting the command-line interface.

2 Command format

- Type one command per line. Command names and field names are lower case.
- Write a value in double quotes, as in `text="Buy milk"`, with no space around `=`. Inside a value, write `\"` for a quote and `\\` for a backslash. Text values cannot be empty.
- Write a time as `yyyy-MM-dd HH:mm`, for example `2026-11-18 16:00`. The date must exist.
- Each record has a positive whole-number ID, shown when it is created. IDs are not reused after a record is deleted.

3 Record types

Type	Fields (all required)	Rule
`note`	`text`	
`task`	`description`, `deadline`	
`event`	`description`, `start`, `alarm`	The alarm must not be later than the start.
`contact`	`name`, `address`, `mobile`	The mobile number is stored exactly as typed.

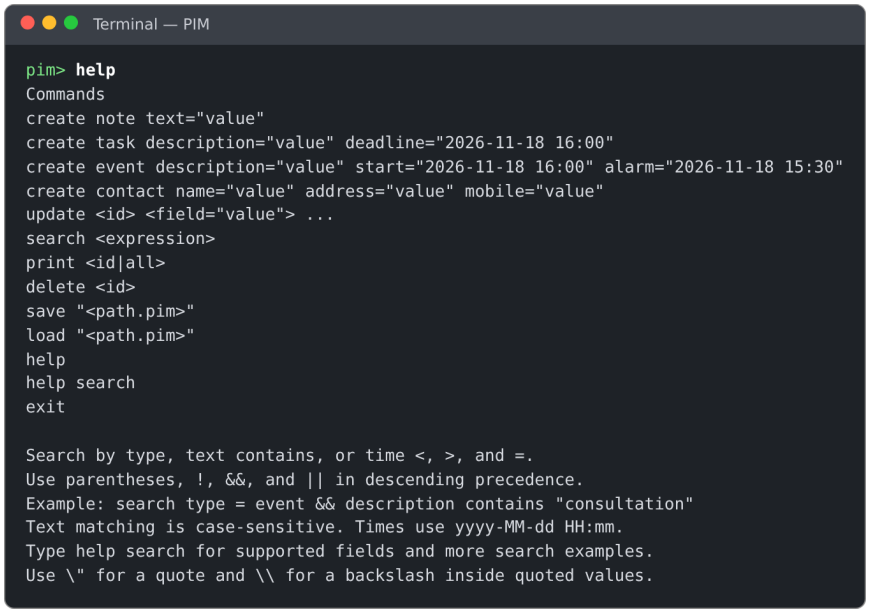


Figure 2: The `help` command lists every command and the search rules.

4 Manage records

Command	Effect
`create note text="Review the SRS"`	Adds a record and prints its ID. Give every field of the type.
`update 1 text="Review the design"`	Changes only the fields given. The whole change is rejected if any value is invalid.
`print 1` `print all`	Shows one record, or all records in ID order.
`delete 1`	Removes the record.

```
create task description="Submit SRS" deadline="2026-11-20 20:00"
create event description="Consultation" start="2026-11-18 16:00" alarm="2026-11-18 15:30"
create contact name="Alex Chan" address="12 Hung Hom Road" mobile="+85212345678"
```

Figure 3 shows records being created, updated, printed and deleted. PIM prints the ID of every new record.

5 Search records

Use `search` followed by one expression. Matching records are listed in ID order, or PIM prints `No matching records.`

- **Type:** `type = note` (also `task`, `event`, `contact`).
- **Text:** `<field> contains "string"` for `text`, `description`, `name`, `address` or `mobile`. Matching is case-sensitive.
- **Time:** `<field> < "time"`, `>` or `=` for `deadline`, `start` or `alarm`.
- **Combine** with `!`, `&&`, `||` and parentheses. The order of precedence is `!`, then `&&`, then `||`.
- A condition on a field that a record does not have is false. To search with `!`, add a type check, for example `type = task && !(description contains "done")`.

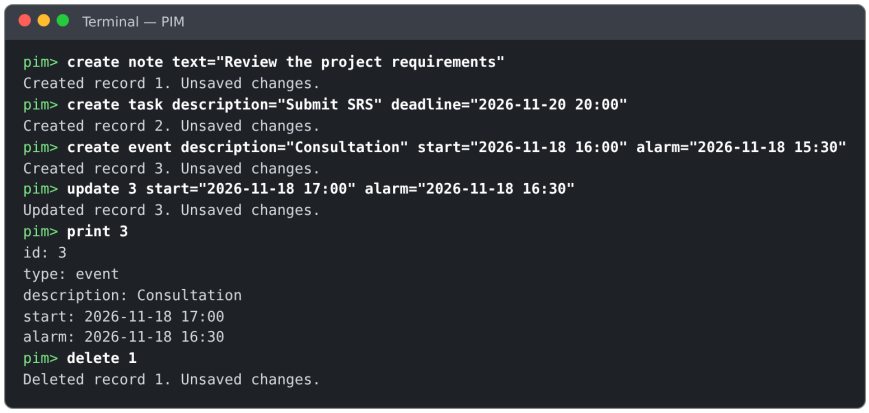


Figure 3: Creating, updating, printing and deleting records.

```
search type = note && text contains "project"
search type = task && deadline < "2026-11-21 00:00"
search (type = task || type = event) && !(description contains "done")
```

Type `help search` for the same rules inside PIM.

Figure 4 shows searches by type, text and time. A search with no result prints `No matching records.`

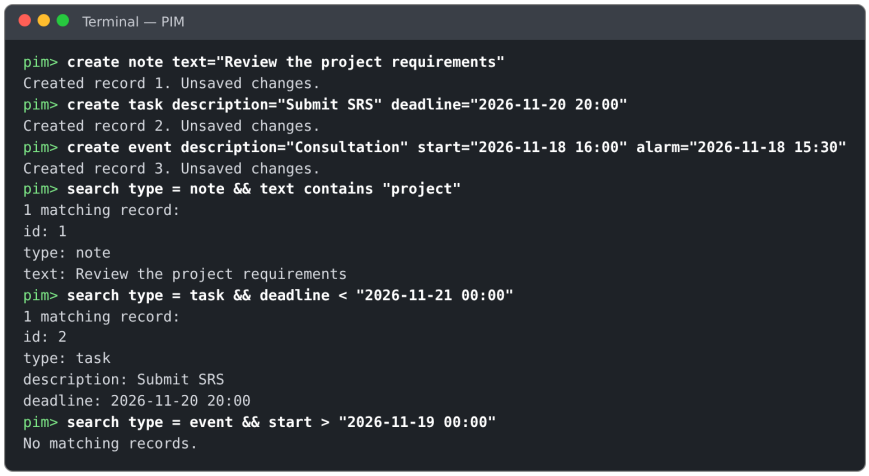


Figure 4: Searching records.

6 Save and load

```
save "my-records.pim"
load "my-records.pim"
```

The file name must be in quotes and end in `.pim`. Relative names are resolved from the folder where PIM was started. PIM asks before overwriting a file and before `load` or `exit` discards unsaved changes. Answer `y` or `yes` to continue; any other answer cancels. `load` replaces all current records, and a missing or invalid file leaves them unchanged.

Figure 5 shows a save, a cancelled load, two rejected commands and a confirmed exit.

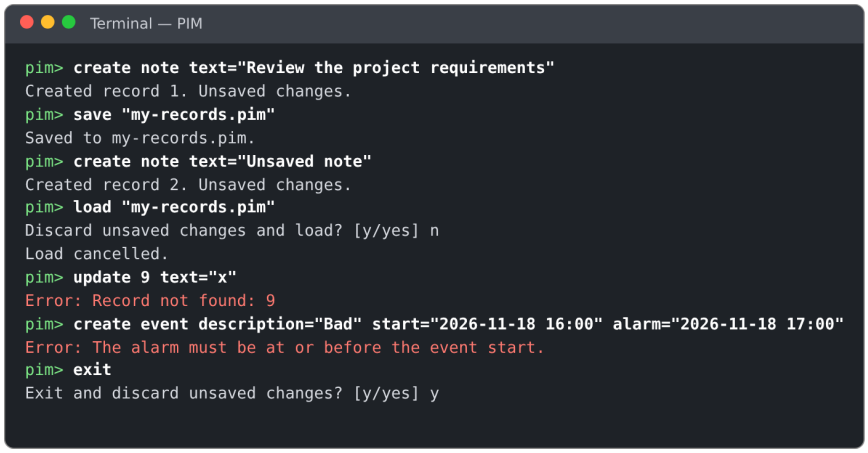


Figure 5: Saving, loading, error messages and exiting.

7 Error messages

A rejected command prints a line starting with `Error:` and changes nothing, so correct the command and type it again.

- `Unknown command` or `Unknown or read-only field`: check the spelling; use lower case.
- `Missing field`, `Duplicate field` or `Field ... must not be blank`: give each field of the record type once, with text.
- `Time must use the format yyyy-MM-dd HH:mm` or `Invalid date or time`: correct the time.
- `The alarm must be at or before the event start`: choose an earlier alarm.
- `Record not found`: check the ID with `print all`.
- `Position n: ...`: the search expression is invalid at character n; see section 5.
- `Cannot save` or `Cannot load`: check the file name, folder and permissions.